Kernix 0.10.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

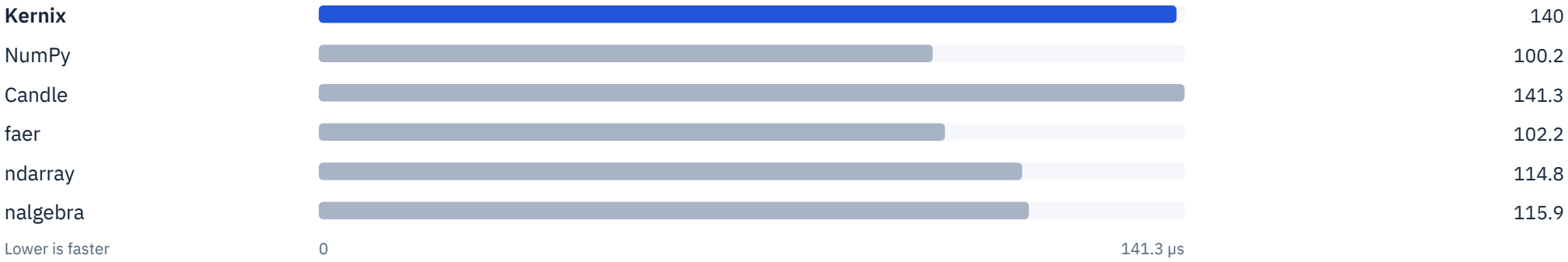
34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



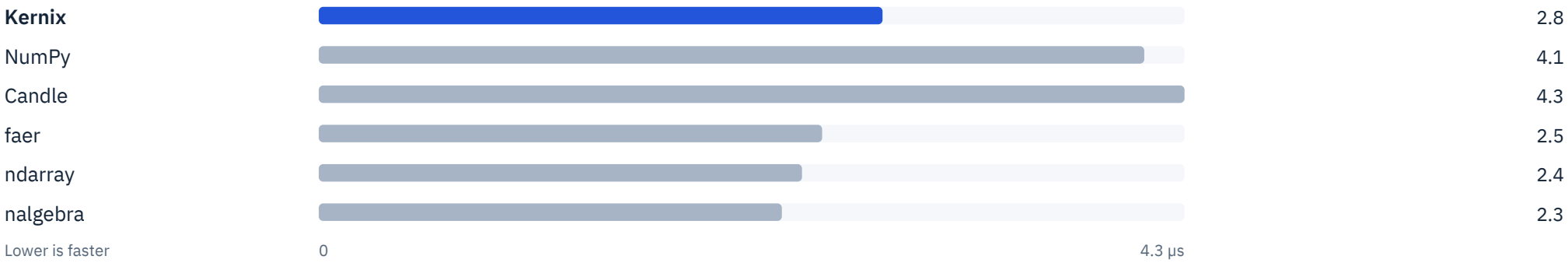
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



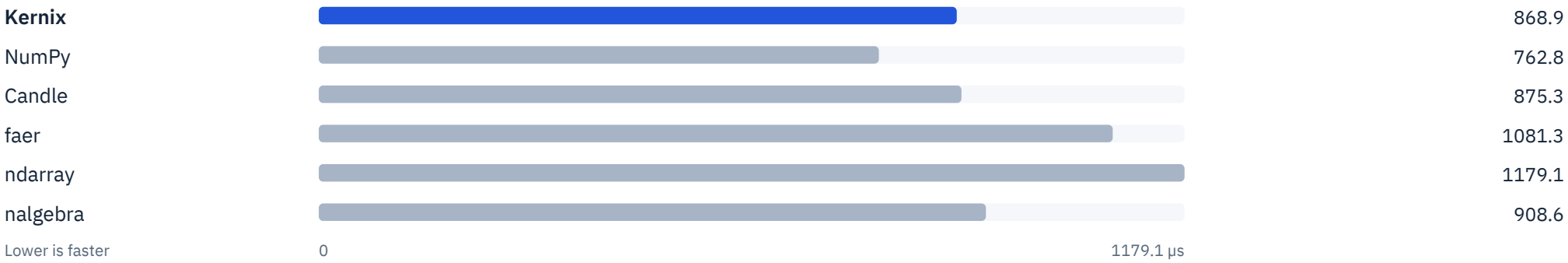
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

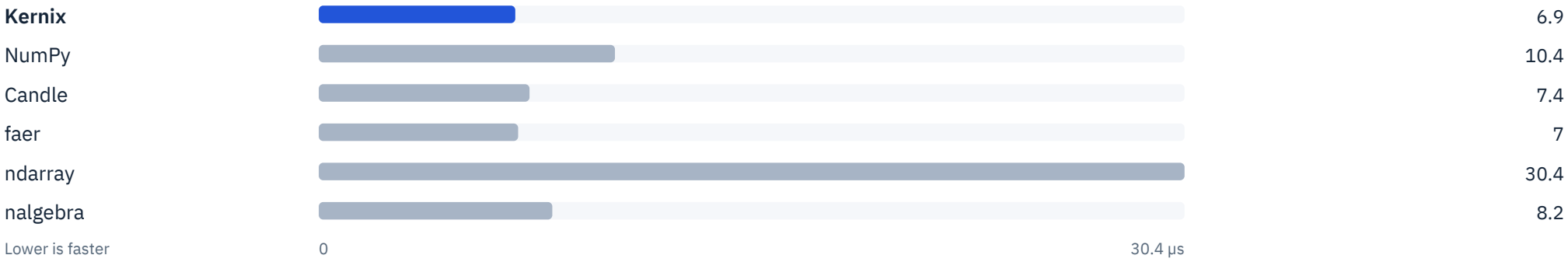
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



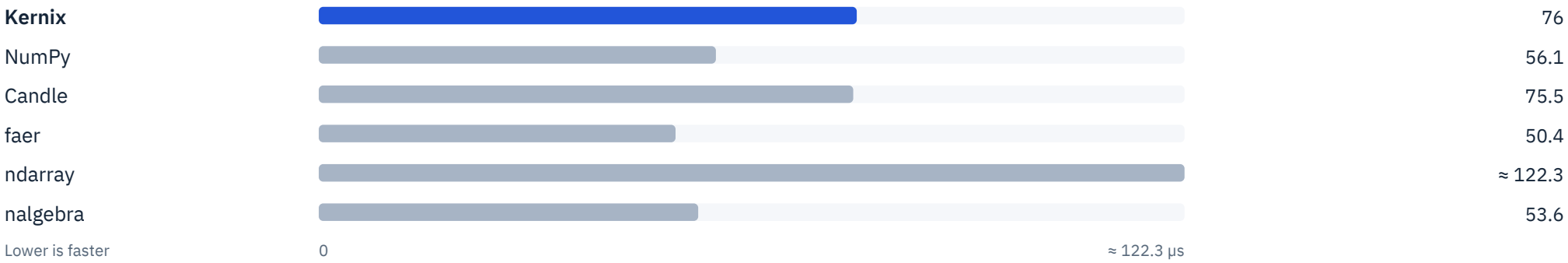
Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

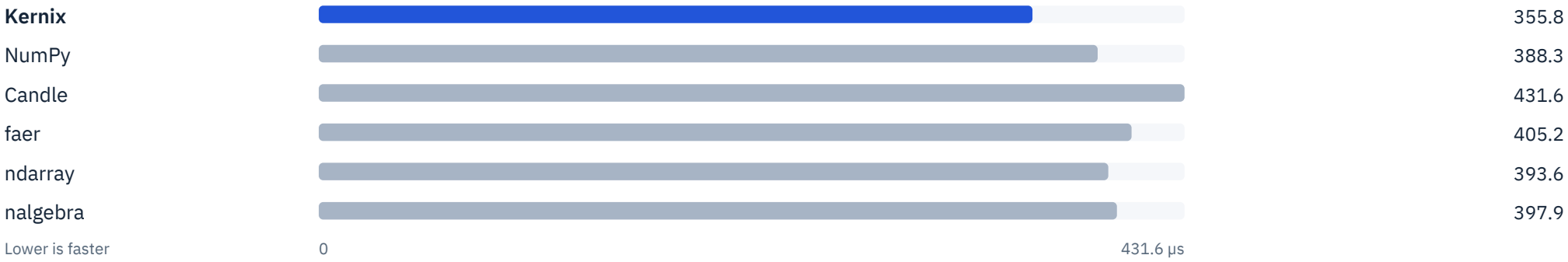
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

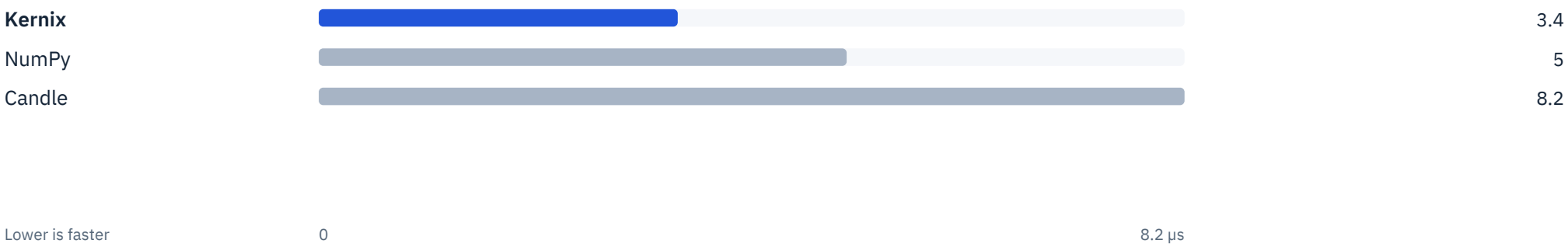


Kernix 0.10.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.10.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded

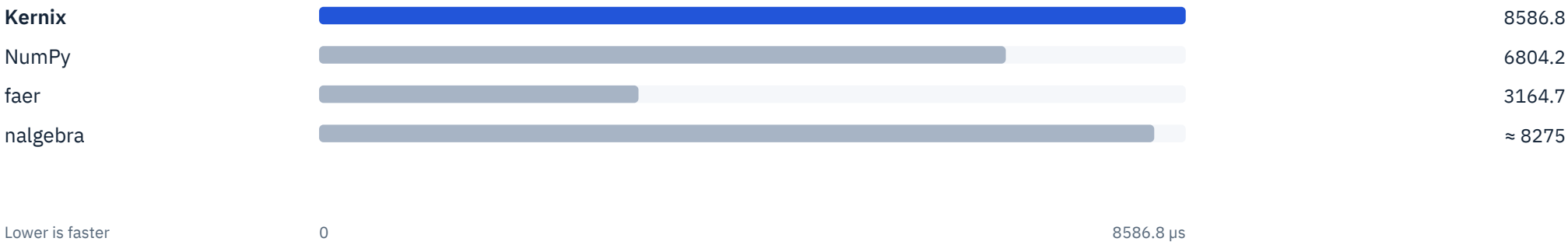


Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

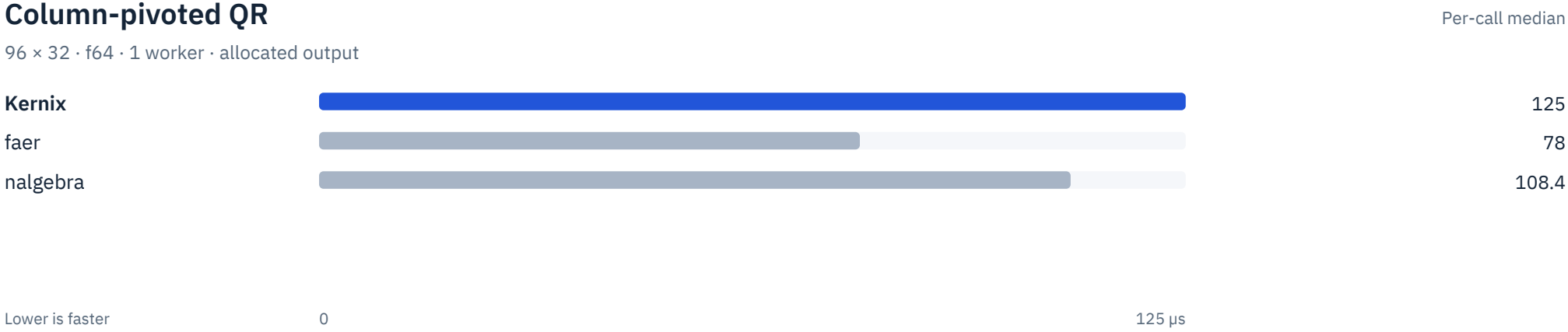
Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



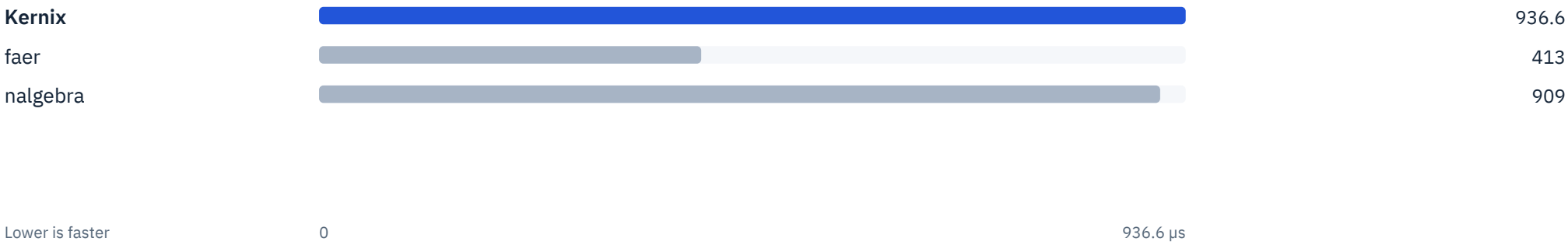
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



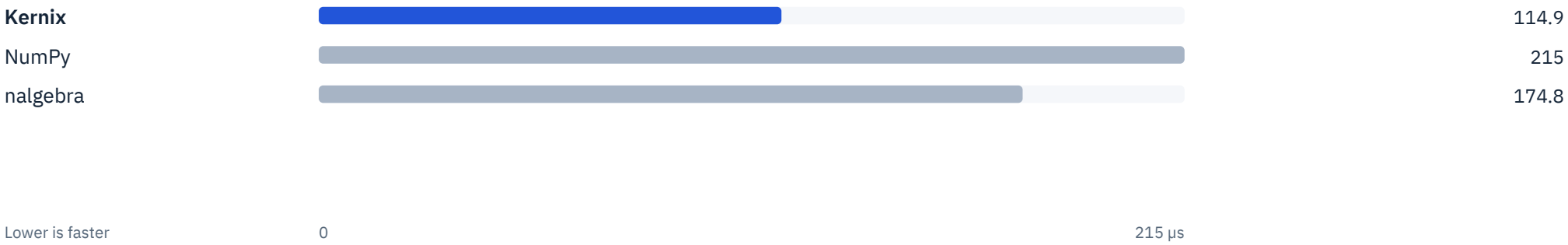
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

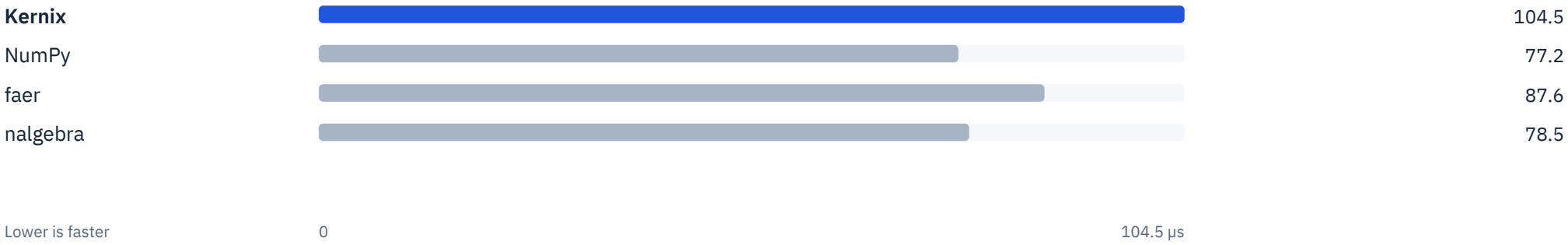
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



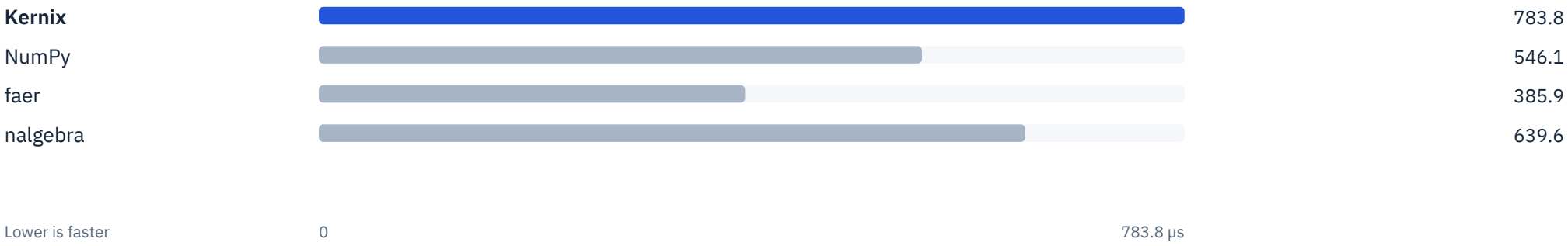
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

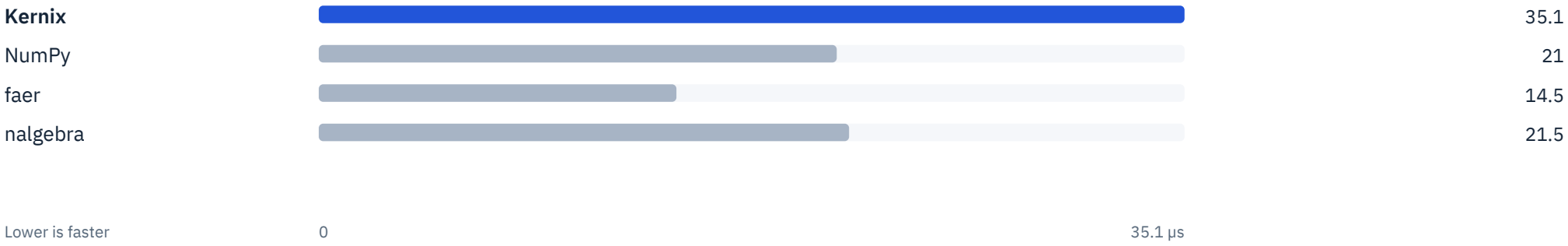
Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



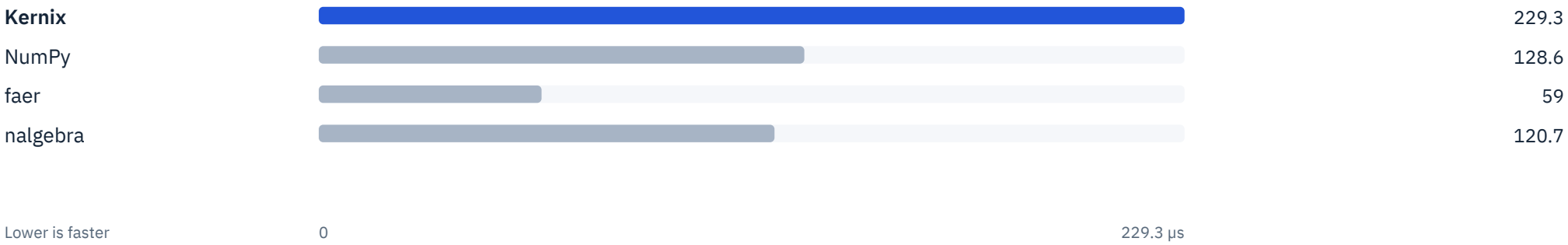
Kernix 0.10.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

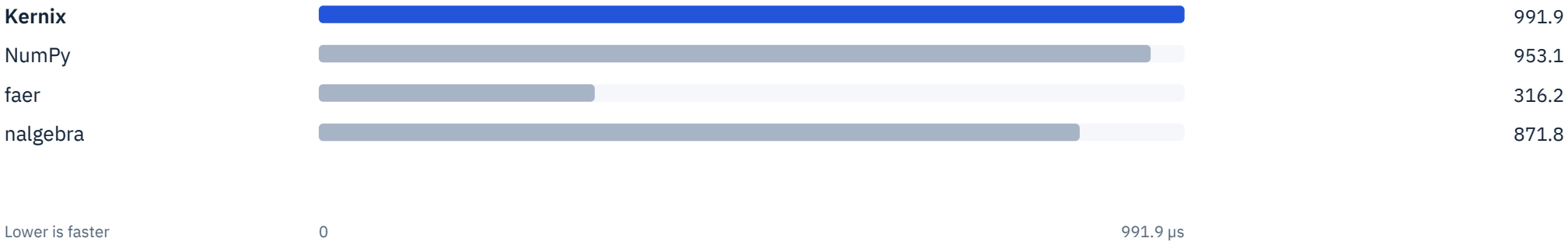
128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

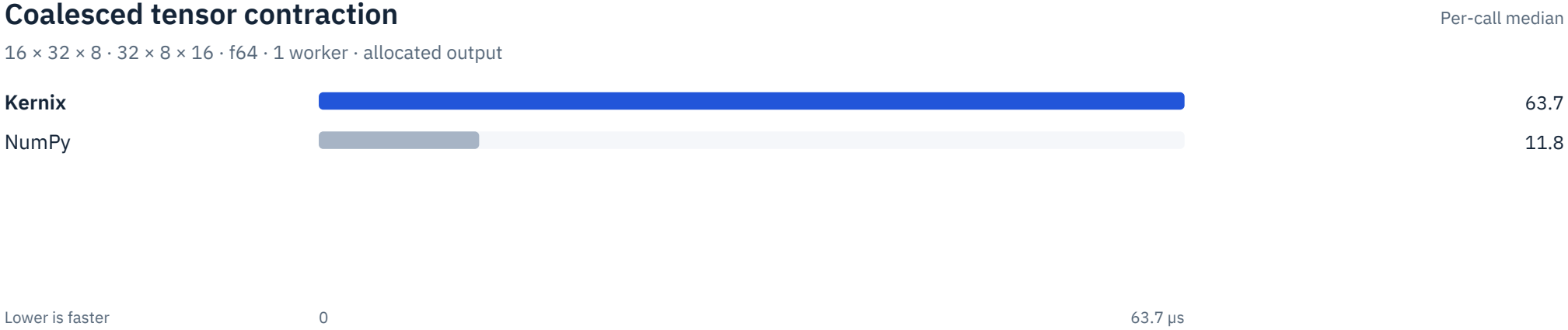


Kernix 0.10.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.10.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



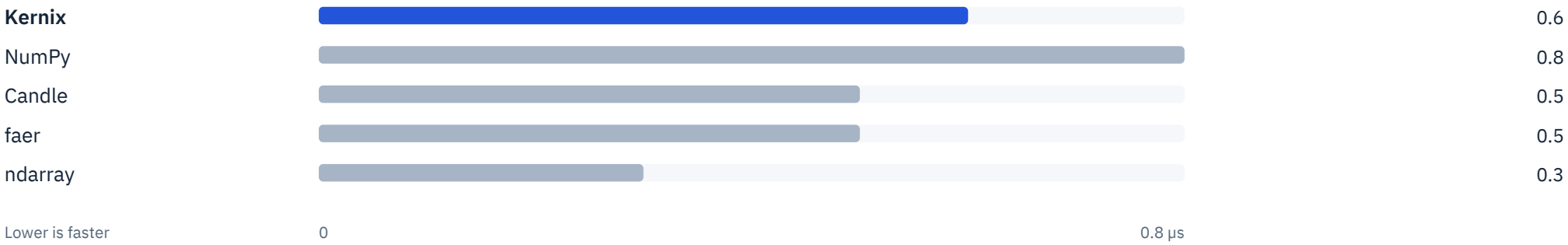
Kernix 0.10.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	141400	141.4
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	102300	102.3
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	56700	56.7
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	75200	75.2
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	4300	4.3
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4100	4.1
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2300	2.3
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2800	2.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	62100.00000000001	62.10000000000001
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14100	14.1
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13600	13.6
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	58500	58.5
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	16000	16
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14700	14.7
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100200	100.2
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	141300	141.3
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	115900	115.9
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	102200	102.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	114800	114.8
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	140000	140
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	868900	868.9
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	762800	762.8
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	908600	908.6
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1179100	1179.1
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	875300	875.3
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1081300	1081.3
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	8200	8.2
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7400	7.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7000	7
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	30400	30.4
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6900	6.9
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	10400	10.4
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	76000	76
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	122299.99999999999	122.29999999999999
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	75500	75.5
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	56100	56.1
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53600	53.6
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50400	50.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	355800	355.8
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	388300	388.3
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	393600	393.6
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	431600	431.6
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	405200	405.2
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	397900	397.9
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	8200	8.2
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	5000	5
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3400	3.4
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	40100	40.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	88400	88.4
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	438700	438.7
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	59000	59
Cholesky factorization / nalgebra 512 × 512 · f64	8274999.999999999	8274.999999999999
Cholesky factorization / NumPy 512 × 512 · f64	6804200	6804.2
Cholesky factorization / faer 512 × 512 · f64	3164700	3164.7
Cholesky factorization / Kernix 512 × 512 · f64	8586800	8586.8
Column-pivoted QR / Kernix 96 × 32 · f64	125000	125
Column-pivoted QR / faer 96 × 32 · f64	78000	78
Column-pivoted QR / nalgebra 96 × 32 · f64	108400	108.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	413000	413
Column-pivoted QR / nalgebra 256 × 64 · f64	909000	909
Column-pivoted QR / Kernix 256 × 64 · f64	936600	936.6
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	104100	104.1
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	114500	114.5
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	100600	100.6
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	174800	174.8
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	215000	215
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	114900	114.9
Thin QR / Kernix 96 × 32 · f64	104500	104.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	87600	87.6
Thin QR / NumPy 96 × 32 · f64	77200	77.2
Thin QR / nalgebra 96 × 32 · f64	78500	78.5
Thin QR / faer 256 × 64 · f64	385900	385.9
Thin QR / Kernix 256 × 64 · f64	783800	783.8
Thin QR / NumPy 256 × 64 · f64	546100	546.1
Thin QR / nalgebra 256 × 64 · f64	639600	639.6
Cholesky factorization / NumPy 64 × 64 · f64	21000	21
Cholesky factorization / nalgebra 64 × 64 · f64	21500	21.5
Cholesky factorization / faer 64 × 64 · f64	14500	14.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	35100	35.1
Cholesky factorization / Kernix 128 × 128 · f64	229300	229.3
Cholesky factorization / nalgebra 128 × 128 · f64	120700	120.7
Cholesky factorization / faer 128 × 128 · f64	59000	59
Cholesky factorization / NumPy 128 × 128 · f64	128600	128.6
Cholesky factorization / faer 256 × 256 · f64	316200	316.2
Cholesky factorization / Kernix 256 × 256 · f64	991900	991.9
Cholesky factorization / nalgebra 256 × 256 · f64	871800	871.8
Cholesky factorization / NumPy 256 × 256 · f64	953100	953.1
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	58600	58.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	9600	9.6
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4400	4.4
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	10100	10.1
Model inference / Candle batched normalized/sliced MLP · f64	10100	10.1
Model inference / Kernix batched normalized/sliced MLP · f64	3500	3.5
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4800	4.8
Model recording + forward / Candle batched normalized/sliced MLP · f64	10200	10.2
Model backward / Kernix prepared graph → fresh gradients · f64	5800	5.8
Model backward / Candle prepared graph → fresh gradients · f64	31900.000000000004	31.900000000000004
Parameter update / Kernix transactional SGD, fixed gradients · f64	1000	1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63700	63.7
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	11800	11.8
Row softmax / Candle 256 × 64 · f64	155800	155.8
Row softmax / ndarray 256 × 64 · f64	153700	153.7
Row softmax / Kernix 256 × 64 · f64	114500	114.5
Row softmax / NumPy 256 × 64 · f64	132600	132.6
Create zeros / Kernix 64 × 64 · f32	600	0.6
Create zeros / faer 64 × 64 · f32	500	0.5
Create zeros / ndarray 64 × 64 · f32	300	0.3
Create zeros / Candle 64 × 64 · f32	500	0.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	800	0.8
Create ones / Candle 64 × 64 · f32	700	0.7
Create ones / NumPy 64 × 64 · f32	2100	2.1
Create ones / Kernix 64 × 64 · f32	500	0.5
Create ones / ndarray 64 × 64 · f32	400	0.4
Create ones / faer 64 × 64 · f32	500	0.5